

Bryson Still

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Research

SMILE Robotics Lab, Clemson University

Robotics Systems Design Engineer

Clemson, SC

May 2026–Present

Advised by Dr. Luyang Zhao, Nature Communications-published researcher in modular soft robotics

- Designed autonomous tool-changer platform with self-aligning drivetrain; enables rapid gripper swap for various grippers on shared base (no calibration); ISO 9409-1 compliant (Flexiv Rizon, Boston Dynamics Spot). Delivers either direct rotary drive or tendon-driven actuation through a single standardized interface
- Built suite of interchangeable end-effectors for the tool-changer platform: custom parallel-jaw gripper with removable adapters (passive-object clamping inserts & slide-on fin-ray finger set), cable-driven continuum end-effector, and a thin-profile gripper (extracts densely-packed objects for Industrial environments).
- Deployed a situational-awareness neural network for autonomous gripper selection, leveraging Qwen 3.5 2B-Instruct and Base language models alongside NVIDIA Cosmos-Reason1 2B vision-language reasoning
- Co-authoring peer-reviewed paper with Dr. Zhao on modular end-effector architecture for reconfigurable manipulation platforms; Cost-optimized production BOM reducing per-unit component cost to sub-\$250.
- Conducted extensive literature review of robotic biomimetic hands (CRAFT, LEAP, Allegro, RUKA, Aero, IIT SoftHand) benchmarked against the 33-grasp Yale GRASP taxonomy, and designed modular biomimetic human-hand prototype with adjustable finger lengths and swappable fingertips tuned to specialty grasps.

Boeing Aerospace

Manufacturing Robotics Intern

Orangeburg, SC

May–August 2024

- Designed a high-precision robotic end-effector for the Alum-A-Lift mobile platform to automate the transport of Boeing 737 lipskins, reducing handling risks for components valued at over \$200,000.
- Engineered custom circuitry and power-management systems for a functional prototype, integrating miniature linear actuators and motor controllers to replicate complex rotational and linear motion.
- Developed 20+ iterative CAD assemblies in SOLIDWORKS and conducted 60+ finite element analysis simulations to validate structural safety margins and predict displacement under maximum operational loads.
- Fabricated a 1/10-scale verification model using a Sermoon D3 Pro 3D printer and performed kinematic testing to confirm that the dual-lever clamping mechanism met IML/OML clearance requirements.

Center for Human-AI Interaction, Collaboration, and Teaming (CU-CHAI)

Undergraduate Engineering Researcher

Clemson, SC

August 2025–Present

- Co-authored a research paper entitled “A Parrot on the Shoulder: Designing AI Mentors with Human Perceptions” (pending review and publication in JHCI).
- Conducted behavioral interviews to collect data on perceptions of human–AI collaboration, informing the development of improved AI-agent architectures. Translated qualitative interview findings into design requirements for AI mentors.

Clemson University Department of Mechanical Engineering

Aerospace Propulsion Researcher, NASA Glenn Research Center Funded

Clemson, SC

June–August 2025

- Conducted aerospace propulsion research under Dr. Yuhao Xu, investigating Leidenfrost droplet dynamics in renewable fuels to characterize heat-transfer behavior, propulsion-cycle efficiency, and combustion emissions.
- Operated an optical data-acquisition system incorporating high-speed imaging and thermal probes to capture droplet evaporation, motion, and surface interactions across 150+ experimental conditions.
- Developed C++ image-processing software and MATLAB analysis workflows to extract quantitative kinematic and thermal data from raw video, reducing manual validation during iterative test campaigns.

Leadership Positions

Clemson CubeSat (University Nanosatellite Program)

Founding Member, ADCS/GNSS Lead

Clemson, SC
Nov 2025–Present

- Co-founded a student-led, student-operated CubeSat team from the ground up,
- Co-authored the AEGISSat-1 mission concept and research proposal, securing acceptance and funding through the Air Force Research Laboratory & University Nanosatellite Program.
- Developed 30+ mission, subsystem, and payload requirements across ADCS, GNSS, and EPS.
- Defined mission objectives, and flight constraints for experiments measuring graphene degradation, electrical resistance, structural-battery cycling, energy capacity, and impedance under atomic-oxygen and ultraviolet exposure.
- Enabled the design, documentation, and technical knowledge of AEGISSat-1's Mission Development to remain publicly accessible for future student teams and collaborators.

Projects

Tractor Retrofit — Engineered retrofit for front loader hydraulic system of JOHN DEERE 5055E tractor to remotely actuate mounted pump. Implemented custom power distribution circuit for actuation in high-vibration agricultural environments, enabling small local farm to protect crops from drought.

MIT IQuhack 2026 (Quantum Computing Hackathon) — Developed animated quantum simulations modeling CZ gate operations; placed within top 4 teams of international competition with 1000+ competitors.

Education

Clemson University

B.S. Electrical Engineering

(Sophomore Standing) May 2029

GPA: 4.0

Relevant Coursework: Machine Design (SOLIDWORKS), Python, Circuits, MATLAB, Digital Logic Design, C & Unix Programming.

Breakthrough Scholars Program — Most prestigious merit scholarship offered at Clemson University, provides full cost-of-attendance for 4 select STEM undergraduates annually.

Technical Skills

Mechanical Design: SOLIDWORKS, Rapid Prototyping, Kinematic Modeling.

Electrical: Electronic System Design, Hardware-in-the-Loop Testing, Oscilloscopes.

Software & Programming: Python, MATLAB, C, Unix/Linux Environments, LaTeX.

Data Collection & Communication: Experimental Data Modeling, Technical Writing (Grant Proposals, Research Papers), Environmental Testing, Performance Simulation.

Publications & Presentations

“A Parrot on the Shoulder”: Designing AI Mentors with Human Perceptions (pending review & publication: International Journal of Human-Computer Interaction) — *2nd Author*

Experimental Investigation of the Leidenfrost Effect in Heptane, Iso-butanol, and Emulsified Oil Droplets — 2025 EUREKA! Poster Forum — *Lead Presenter*

Datasets & Repositories

Pongo pygmaeus hand bone STL meshes: CT-derived surface models of an adult male Bornean orangutan Still, B. (2026). *MorphoSource*, [ark:/87602/m4/883941](https://morphosource.org/ark:/87602/m4/883941). Derived from AMNH:Mammals:M-28253, Media 000075137.

- Open-Sourced the first STL of the American Museum of Natural History's CT scan of an adult male Bornean orangutan hand, formally accepted into the AMNH Mammal Collections repository. Utilized and Refined a morphological smoothing algorithm & 3D-Slicer to generate watertight, 3D-printable meshes. Currently serves as the geometric foundation for an independent biomimetic robotics research project on tendon-driven arboreal grippers.